

server and an application server on this physical server. You need to make a choice between deploying separate virtual machines for the Web server and the app server, and hosting them in two different Docker containers. What would you choose and can you explain the rationale behind your choice?

8. In Problem 7, you are now told that you don't need to host an application server, but you need to host as many Web servers as possible on the physical machine. Would you now use a virtual machine to host each Web server instance or would you use a Docker container for each Web server? Can you explain the reason for your choice?
9. Most of you would be familiar with cloud computing services such as Amazon EC2. Amazon also provides cloud storage through Amazon Simple Storage, popularly known as Amazon S3. This object based cloud storage service can be accessed via HTTP services, and your files can be stored as objects in it. What are the advantages of an object based storage service as opposed to a file based storage service?
10. Consider Question 9 and imagine you are running a database server instance on Amazon EC2. How would you host your application data on the cloud? If you are asked to choose between object based cloud storage and block based cloud storage, what would you pick? Can you provide the reasons for your choice?
11. Assume that you have been asked to move your on-premise enterprise payroll application onto the cloud. You have decided on the cloud compute and storage vendor. Your application has 20TB of data, which needs to be hosted onto the cloud. How would you ensure the movement of this data from your on-premise storage to the cloud? What are the issues and precautions that you need to take?
12. Consider that you have stored your critical enterprise data on cloud storage. How do you protect against any data loss due to a failure of the cloud storage system?
13. We are all familiar with the use of RAID in enterprise storage for redundancy so as to avoid data loss. There are two ways of preventing data loss in enterprise storage. One is to have multiple copies of the data. The other is to store the data in erasure coded format so that in the event of a disk failure, the data can be reconstructed from the erasure coded format. What are the pros and cons of these two choices? For cloud storage, would you opt for storing the data in erasure coded format or by keeping multiple copies of the data? Explain the reasons for your choice.
14. Your organisation has been asked to evaluate how using cloud compute and storage compares with using on-premise compute and storage for its IT needs. Which of your enterprise applications would you consider moving to the cloud and why?
15. Assume that you have moved your organisation's payroll data into the cloud. However, your organisation has four offices and all of them could be updating the payroll database at the same time. How does the cloud storage system synchronise these multiple updates from different geographical locations onto the same data object? While traditional database or file locking would take care of this issue in on-premise data storage, can you use the same mechanism to ensure synchronised, ordered updates from multiple geographical locations onto the same data object? If yes, explain how? If not, explain why traditional file or database locking mechanisms would not serve the purpose?
16. Your enterprise applications have widely varying compute demands, over a period of time. For instance, during month-ends, there is excessive demand for compute to run batch jobs. Similarly, for a couple of hours each night, there is a spike in compute resource usage, again due to batch jobs. How would you handle such dynamic resource demand variations if you need to move your enterprise application onto the cloud?
17. Assume that data redundancy on cloud storage is achieved by means of maintaining three copies of each piece of data. When a user-initiated data change occurs, how are the three copies updated? Would the user-initiated write return success to the user only after all the three copies are updated? Or would it return success right after updating any one of the copies successfully? What are the pros and cons of these two choices?
18. Assume that you have bought cloud compute and cloud storage and have moved your enterprise application onto the cloud. Consider two scenarios—one in which the same vendor provides both compute and storage on the cloud, and another in which the cloud compute and cloud storage come from two different vendors. In each of these scenarios, how do you ensure the locality of data for your cloud hosted application?

If you have any favourite programming questions or software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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